

Literature-Implied Answer: Positive UMD-Lattice Semigroups Need Not Be $\mathcal{R}$ -Sectorial

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Status

This packet records a *literature-implied* negative answer to the positivity-only UMD-Banach-lattice version suggested by the open-problem cluster in Fackler's survey arXiv:1407.1142. It is not claimed as a new counterexample and it is not an answer to the contractive variants.

The source problem cluster asks whether positivity and/or contractivity of the semigroup generated by $-A$ force bounded H^∞ -calculus, bounded imaginary powers, or $\mathcal{R}$ -analyticity.

some open questions regarding generalizations of Weis' result.

Problem 2. Let A be a sectorial operator on some UMD-Banach lattice and suppose that $-A$ is the generator of a positive contractive C_0 -semigroup. Does A have a bounded H^∞ -calculus (bounded imaginary powers / is $\mathcal{R}$ -analytic)?

Problem 3. Let A be a sectorial operator on some L_p -space for $p \in (1, \infty)$ and suppose that $-A$ is the generator of a contractive C_0 -semigroup. Does A have a bounded H^∞ -calculus (bounded imaginary powers / is $\mathcal{R}$ -analytic)?

Problem 4. Let A be a sectorial operator on some L_p -space for $p \in (1, \infty)$ and suppose that $-A$ is the generator of a positive C_0 -semigroup. Does A have a bounded H^∞ -calculus (bounded imaginary powers / is $\mathcal{R}$ -analytic)?

Supporting Literature

Fackler's later paper arXiv:1411.4240 proves the following theorem.

With these preliminary observations we obtain the following result.

Theorem 3.3. *Let X be a Banach space that admits a normalized non-symmetric 1-unconditional Schauder basis $(e_m)_{m \in \mathbb{N}}$. We consider X as a Banach lattice with the order induced by $(e_m)_{m \in \mathbb{N}}$. Then there exists a non- $\mathcal{R}$ -sectorial operator A with $\omega(A) = 0$ such that $-A$ generates a positive analytic C_0 -semigroup on X .*

Proof. It follows from [Sin70, first part of the proof of Proposition 23.2] that there

The same paper immediately specializes the theorem to the UMD Banach lattices $\ell_p(\ell_q)$, $p, q \in (1, \infty)$, $p \neq q$.

exists in $\text{Rad}(X)$. By [DJT95, Theorem 12.3] and the unconditionality of the basis, $\sum_{m=1}^{\infty} a_m e_{2m-1}$ converges. This contradicts the choice of $(a_m)_{m \in \mathbb{N}}$ and therefore A is not $\mathcal{R}$ -sectorial. $\square$

We now give some concrete examples of spaces for which the above theorem can be applied.

Example 3.4. For $p, q \in (1, \infty)$ consider the UMD-spaces $\ell_p(\ell_q)$ with their natural lattice structure. Its ordering is induced by the standard unit vector basis $(e_m)_{m \in \mathbb{N}}$ of $\ell_p(\ell_q)$ for some enumeration of $\mathbb{N} \times \mathbb{N}$. Clearly, $\ell_p(\ell_q)$ contains both copies of ℓ_p and ℓ_q and therefore for $p \neq q$ the basis $(e_m)_{m \in \mathbb{N}}$ is 1-unconditional and non-symmetric. Hence for $p \neq q$, Theorem 3.3 yields a sectorial operator A on $\ell_p(\ell_q)$

with $\omega(A) = 0$ such that $-A$ generates a positive analytic C_0 -semigroup without maximal regularity.

Implication

Choose $p, q \in (1, \infty)$ with $p \neq q$. The space $\ell_p(\ell_q)$, with its natural lattice structure, is a UMD Banach lattice. By Theorem 3.3 and Example 3.4 of arXiv:1411.4240, there exists a sectorial operator A on $\ell_p(\ell_q)$ such that

$$\omega(A) = 0, \quad -A \text{ generates a positive analytic } C_0\text{-semigroup,}$$

but A is not $\mathcal{R}$ -sectorial.

The survey arXiv:1407.1142 recalls the implication chain relevant here: bounded H^∞ -calculus implies bounded imaginary powers, and on UMD spaces bounded imaginary powers imply $\mathcal{R}$ -sectoriality. Therefore the above operator A cannot have bounded imaginary powers and cannot have a bounded H^∞ -calculus. Since it is not $\mathcal{R}$ -sectorial, it is also not $\mathcal{R}$ -analytic in the sense of the source problem cluster.

Thus positivity alone on UMD Banach lattices does not force the regularity properties asked for in the cluster.

Limitations

- The semigroup in arXiv:1411.4240 is positive analytic, but no contractivity is asserted. Hence this packet does not settle the positive contractive UMD-Banach-lattice problem in arXiv:1407.1142.
- The counterexample space $\ell_p(\ell_q)$, $p \neq q$, is a UMD Banach lattice but not a classical scalar L_p space. Hence this packet does not settle the positive L_p problem or the contractive L_p Matsaev problem.

- The packet records a direct literature implication from Fackler's later theorem; it does not add a new construction beyond that theorem.